

ORIGINAL ARTICLE

A biochemical and histological approach to study antifertility effects of methanol leaf extract of *Asplenium dalhousiae* Hook. in adult male rats

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Abstract

The study was designed to investigate the antifertility properties of methanol leaf extract of *Asplenium dalhousiae* in adult male rats. Forty adult male Sprague Dawley rats (150 ± 10 g) divided into four groups ($n = 10$ animals/group) were administered with different doses (0, 50, 100, 150 mg/kg) of plant extract for 28 days. On day 29th, rats were decapitated, trunk blood and reproductive tissues were collected, and blood plasma was separated and stored until use for measuring reproductive hormones, while epididymis and testis were used for assessment of sperm parameters, oxidative stress status and morphometric analysis. Sperm motility, viability and sperm production rates were lowered in high dose treatment groups. Levels of catalase (CAT), sodium dismutase (SOD) and peroxidase (POD) decreased while stress biomarkers including reactive oxygen species (ROS) and thiobarbituric acid reactive substances (TBARS) were increased among all treatment groups. Concentrations of plasma testosterone and follicle stimulating hormone (FSH) were decreased while levels of luteinizing hormone (LH) increased in high extract treated groups. Histological examination of testis showed disorganisation of seminiferous tubule and reduced spermatocytes number. The findings of current study revealed that methanol leaf extract of *A. dalhousiae* might induce antifertility effects via oxidative stress and interfering with testicular architecture leading to spermatogenic arrest.

KEYWORDS

Asplenium dalhousiae, biochemical analysis, herbal drugs, histological analysis, male contraceptives

1 | INTRODUCTION

Since olden times, methods of family planning have focused chiefly on use of female contraceptives, while those for men gained little attention. Today, the majority of contraceptives available in markets are women oriented (Lampiao, 2011). Contraceptive practices available for men consist of condoms, vasectomy and withdrawal of the penis from the vagina prior to ejaculation during sexual intercourse (Bremner, 2012). Recently, clinical research in male

contraception has generally focused on hormonal approaches (Page, Amory, & Bremner, 2008), although such approaches have shown effective antifertility potential with failure rates much lower than that of condoms (Bremner, 2012). Still, hormonal contraceptives are reported to have carcinogenic effects (Devi, Kumar, Nidhi, & Dhamija, 2015). Therefore, these days plant products catch the attention of many scientists as a primary source of naturally occurring fertility regulating agents because of their little and no side effects.

Several plant products possess antifertility activities and may be developed into male contraceptives (Devi et al., 2015; Joshi, Sharma, & Chaturvedi, 2011). Numerous plants containing components like flavonoids, terpenes, tannins, quinines and diterpenoid lactones have been described to inhibit male fertility suppression properties by different mechanism (Joshi et al., 2011; Koteswara Rao, Vimalamma, Venkata Rao, & Tzeng, 2004; Reddy et al., 2003). A variety of plants have been reported to hinder different stages of spermatogenesis in many animals like rats, mice, dogs, humans and monkey (D'Cruz, Vaithinathan, Jubendradass, & Mathur, 2010; Dixit, Khanna, & Bhargava, 1978; Jahan et al., 2009; Revathi, Vani, & Sarathchandiran, 2010).

Herbal medicines and their derivatives have been incorporated into traditional medicine virtually since the beginning of recorded history (Daniyal & Akram, 2015). Studies of plant components with contraceptive characteristics represent possible alternatives to birth control in the available methods. In addition, plants can be used as antifertility agents, and these can be classified according to their activity characteristics, such as anti-fertilisation plants; spermicidal and semen coagulating plants; and growth-suppressing plants (Singh et al., 2013). Compounds from plant like lupeol in mangoes and pristimerin compounds could block the progesterone activating cation channels of spermatozoon (CatSper) and affected sperm fertility. Pristimerin and lupeol can act as contraceptive compounds by averting sperm hyperactivation, thus preventing fertilisation (Mannowetz, Miller, & Lishko, 2017).

The genus *Asplenium* of the family Aspleniaceae is one of the largest fern genera in Europe, consisting of 30 species previously reported (Reichstein, 1981). Recently, it has been reported to have 700 species (Ajay Singh Bisht, 2015). *Asplenium dalhousiae* Hook. (the plant name has been verified from <http://www.theplantlist.org>), commonly known as "Haspa Buty" or "maiden spleenwort," is a medium sized herb with a bushy stem. The plant is also known by other synonyms that are *Asplenium rupium* Goodd., *Ceterach alternans* Kuhn, *Ceterach dalhousiae* (Hook.) C. Chr. and *Ceterachopsis dalhousiae* (Hook.) Ching (<http://www.theplantlist.org>). In Pakistan, it is widely distributed in District Shangla and its nearby areas Alpuri, Shahpur, Lilowni, Chakesar, Hayatabad, Swat, Murree, Dir, Ajaori and Kass. The fern is useful for ailments of spleen. Decoction of whole plant is antiviral and also useful to cure skin blisters, and aqueous extract of its fronds is given in snake bite (Chander, Choudhary, & Sharma, 2017; Khan, Khan, Hussain, & Mujtaba, 2012). It is also used as antihelmintic and vermifuge. The traditional use in Indo-Pakistan region is that the decoction prepared from roots of *A. dalhousiae* is given to infants as "gutti." The plant leaves have been known to be locally used by Pakistani tribes of Dir, Balakot, Changla gali and Abbottabad for antifertility purposes. The extraction of the plant leaves is used before breakfast for 15 days in the form of infusion for contraceptive purpose in men (Zaman et al., 2018).

Physicochemical studies have shown that the leaf of the plant contains total moisture content 12%, total ash 1%, total starch content 0.13%, total sugar content 0.19%, total tannins 1.18%, total phenolics 9.27%, total flavonoids 1.3% and total

proanthocynadines content 0.01% (Ajay Singh Bisht, 2015). *Asplenium dalhousiae* contains many phytochemicals like Flavanol (Kaempferol 3-glucuronide, Kaempferol 3-digalactoside, Quercetin 3-glucuronide, Quercetin 3-digalactoside, Kaempferol 3-(6-malonylglucoside), Quercetin 3-glucoside, Quercetin 3-gentiobioside, Quercetin 3-(6-alonylgalactoside), Flavanone (Naringenin 7-neohesperidoside); Zaman et al., 2018). Flavanol, Kaempferol and Quercetin are used as antifertility agents in males (Daniyal & Akram, 2015). Thus, due to presence of flavanoids, kaempferol and quercetin in our study plant it is inferred that *A. dalhousiae* leaf extract can act as an antifertility agent. Therefore, this study was designed to evaluate the fertility suppressing properties of *A. dalhousiae* leaf extract by targeting process of spermatogenesis and testicular damage if any, in adult male rats.

2 | MATERIALS AND METHODS

2.1 | Plant collection and extract preparation

Asplenium dalhousiae leaves were obtained from agricultural fields of Dir, Pakistan, at elevation of 1,420 m. The plant species was identified and authenticated by Dr. Mushtaq Ahmad, Associate Professor, Department of Plant Sciences, and sample was deposited in the herbarium of Plant Sciences department, QAU, under accession number 105632.

The leaves were air-dried for about 2 weeks, ground in Waring blender and then sieved through filter paper. The dried powder was extracted with methanol (leaf to solvent ratio 1:10) followed by filtration using Whatman™ Qualitative Filter Paper Standard Grade, circle whose pore size was 90 mm (GE Healthcare 1001090 Whatman™ Qualitative Filter Paper: Grade 1 Circles, Thermo Fisher Scientific) and then concentrated on a rotary evaporator (Model: Hei-VAP Heidolph, Germany; Gulfranz, Qadir, Nosheen, & Parveen, 2007).

2.2 | Animals

Forty adult male *Sprague Dawley* rats with an average weight of 150 ± 10 g were used. Animals were housed in separate stainless steel cages in well-ventilated rooms with temperature of 20–26°C and 12 hr dark/light cycle for 28 days. Standard laboratory pellet food and tap water was provided to animals ad libitum throughout experiment. All the animal handling and procedures were approved by the Ethical Committee of Animal Sciences Department.

2.3 | Experimental design

The animals were randomly placed in four different groups with ten animals in each group. Group I served as control group, and Groups II, III and IV were treated with different methanol extract (ME) doses 0 mg/kg (Control), 50 mg/kg (ME1), 100 mg/kg (ME2) and 150 mg/kg (ME3) body weights of leaf extract by oral route for 28 consecutive days.

2.4 | Animal dissection and tissue collection

At day 29, animals were weighed and decapitated. Following decapitation, trunk blood was collected directly in heparinised syringes and centrifuged at 1008 g for 15 min. Plasma was separated and kept at -20°C until hormonal analysis. Both testicular and epididymal tissues were dissected out from all the animals, followed by washing in saline and then weighed. Left testis was washed in ice cold saline and stored in freezer (-20°C) for analysis of daily sperm production and biochemical determination. Left epididymis was washed, weighed and then minced for further assessment of epididymal sperm count and other sperm parameters. Right testicular tissue was fixed in 10% formalin for histological studies.

2.5 | The body and accessory reproductive organs weights

The body weights of adult rats were recorded before and after the completion of experiment. All the reproductive organs including testis, epididymis, seminal vesicle and prostate gland were taken out, trimmed free of fat, dried by blotting on filter paper and weighed separately on single pan electronic balance (Electronic compact scale SF-400C).

2.6 | Effect on sperm motility

In order to determine the sperm count and motility, a small part of cauda epididymis was resected and crushed in 1 ml of 37°C normal saline solution. Ten microlitre of this homogenate sample was placed on a pre-warmed slide, a minimum of 10 fields were observed, and 100 spermatozoa were assessed by high power microscopy ($40\times$ magnifications, Nikon Model YS 100; Ain, David, Shah, Ahmad, & Jahan, 2018).

2.7 | Assessment of sperm viability

In order to assess sperm viability, eosin-nigrosin staining test was used. Momentarily, 25 μl of dye (eosin-nigrosin) and the equal amount of semen sample were mixed for 30 s. Then, smear was prepared on a glass slide by a 15 μl droplet of the mixture. After drying the smear at room temperature, the slides were examined under a light microscope (Nikon Model YS 100, $40\times$ magnification). Live spermatozoa were unstained (white) but dead cells were stained red. After counting at least 100 sperm cells, the percentage of dead and alive spermatozoa was calculated (Ain et al., 2018).

2.8 | Daily sperm production

The earlier frozen testicular tissues were thawed at room temperature for few minutes, followed by homogenisation in 25 ml of 0.9% sodium chloride (NaCl) containing 0.5% triton X-100, for 30 s. The produced homogenate was diluted 5-fold, 20 μl of sample from the homogenate was deposited on Neubauer's chamber,

and observed under microscope ($40\times$ magnification, Nikon Model YS 100), late spermatids were counted (Ain et al., 2018). To obtain average number of spermatids, three readings were taken twice for each sample. For the calculation of DSP, the number of spermatids which was noneffective against homogenisation was divided by 6.3, which represents the number of days these spermatids resides in seminiferous epithelium calculated by the following formula.

$$Y = \text{No. of sperms (N)} \times 25 \times 1000 \times 5 \times 25 = Y/6.3$$

2.9 | Epididymal sperm count

In order to determine the epididymal sperm count, small part of caput, corpus and cauda epididymis was cut, kept in 1 ml of 37°C normal saline and crushed. Two to three drops of nigrosine were added into it. With the help of a pipette, 10 μl of this homogenate sample was taken and placed on a pre-warmed slide. A minimum of 10 fields were observed, and spermatozoa were counted in each part of epididymis by high power microscopy ($40\times$ magnifications, Nikon Model YS 100; Ain et al., 2018).

2.10 | Biochemical analysis

The testicular samples stored at -20°C were thawed and homogenised in 3 ml of phosphate buffer (pH 7.4). The homogenate was centrifuged at 100,800 g for 30 min at 4°C . Supernatant was collected for determination of antioxidant status of testicular tissue.

Activity of antioxidant enzymes like catalase (CAT), sodium dismutase (SOD), peroxidase (POD) as well as stress biomarkers reactive oxygen species (ROS) and thiobarbituric acid reactive species (TBARS) was estimated in testicular tissue of all the experimental animals by using methods described by Chance and Maehly, (1955), Wright, Colby, and Miles (1981) and Kakkar, Das, and Viswanathan (1984), respectively using Bio Rad SmartSpecTM Plus Spectrophotometer (Chance & Maehly, 1955; Kakkar et al., 1984; Wright et al., 1981).

2.11 | Hormone analysis

Testosterone levels in blood plasma and testis were determined by enzyme immunoassay (EIA), using commercially available kits (Biocheck Inc, USA). Luteinizing hormone (LH) and follicle stimulating hormone (FSH) concentrations were measured through enzyme-linked immunosorbent assay using commercial kits (Reddot Biotech Inc.) using microplate reader (Platos R 496; AMP Diagnostics). All the samples were measured in a single assay. Results were expressed as ng/ml and IU.

2.12 | Tissue histology

Testicular tissues placed in formalin were dehydrated in different grades of alcohol, followed by embedding in paraffin wax. Later, these embedded tissues were mounted on wooden blocks with

the help of melted wax. Seven micrometer thin sections were cut using microtome (Thermo, Shandon finesse 325, UK). The long ribbons with tissues in it were stretched, followed by fixation in clean albumenised glass slides, previously prepared. These slides were kept on Fischer slide warmer (Barnstead Labline series 28026) at 60°C overnight. Next day, these slides were stained and permount with DPX.

2.13 | Light microscopic study

Using light microscope (Nikon, 187842, Japan), 7 µm thick tissue sections were studied at 40× magnification. Slides of all the experimental groups were studied. By using Leica LB microscope (Germany) equipped with cannon digital camera (Japan), microphotography was done. For histomorphometric analysis, different morphology parameters were studied using image J2× software package program.

2.14 | Statistical analysis

All the results were expressed as mean ± SEM. The data were distributed normally, and one-way analysis of variance (ANOVA) followed by post hoc Tukey's test was used for the statistical analysis of data using Graph pad Prism software version 5.00 for Windows, GraphPad Software, San Diego CA, USA, (www.graphpad.com). The significance level was set at $p < 0.05$.

3 | RESULTS

3.1 | Body weight

A prominent increase in body weight was detected in animals of all the leaf extract treated groups in comparison with control group. A significant increase ($p < 0.001$) was also observed when ME 2 group was compared with control, whereas there was no change in body weight seen between control and ME 3 as shown in Table 1.

3.2 | Reproductive organs weight

On comparison of control and ME 1, a prominent decrease ($p < 0.001$) in testicular weight was observed. A highly significant increase in testicular weight was observed between control and ME 3 (Table 1). Similarly, there was a significant increase ($p < 0.001$) observed in weight of epididymis when group ME1 and ME3 were compared to control group. However, no significant difference was noticed in mean weight of prostate and seminal vesicle after treatment with plant extract.

3.3 | Sperm motility and viability

A significant reduction in sperm motility ($p < 0.05$) as well as viability ($p < 0.01$) rate was seen in ME 2 group in comparison with control group. No significant difference was observed in mean per cent (%) motility rate of spermatozoa, when treated groups (ME 1 and ME 3) were compared to control as well as among each other (Table 2).

3.4 | Daily sperm production

A significant decrease ($p < 0.001$) in daily sperm production was noticed in all the treated groups when compared to control (Table 2). Significant reduction in sperm production was also seen when ME 1 and ME 2 groups were compared to ME 3 ($p < 0.05$ and $p < 0.001$ respectively).

3.5 | Epididymal sperm count

Sperm number in caput epididymis was lessened in all treatment groups. However, a significant reduction in ME 1 ($p < 0.01$) and ME 2 ($p < 0.05$) was observed as compared to control group. No significant change was seen in corpus epididymis among control and all the treated groups. A significant decrease in the cauda sperm count was noted in ME 1 when compared to control group (Figure 1).

TABLE 1 Effect of *Asplenium dalhousiae* leaf extract on mean ± SEM body and reproductive organs weight (g) of control and treated adult male rats

Parameters	Treatments			
	Control	ME 1 (50 mg/kg)	ME 2 (100 mg/kg)	ME 3 (150 mg/kg)
Body weight gain (g)	56.50 ± 6.55	91.40 ± 5.72	107.60 ± 3.15 ^{a***}	55.50 ± 4.09
Right testicular weights (g)	1.33 ± 0.04	1.14 ± 0.02 ^{a***}	1.37 ± 0.02	1.26 ± 0.06
Left testicular weights (g)	0.41 ± 0.04	0.55 ± 0.03	0.51 ± 0.02	0.59 ± 0.05 ^{a***}
Epididymis weight (g)	0.88 ± 0.10	1.23 ± 0.06 ^{a***}	1.13 ± 0.06	1.24 ± 0.12 ^{a***}
Seminal vesicle weight (g)	0.83 ± 0.08	1.09 ± 0.09	1.07 ± 0.20	1.25 ± 0.16
Prostate weight (g)	0.46 ± 0.07	0.65 ± 0.04	0.59 ± 0.04	0.64 ± 0.07

Notes. Values are expressed as mean ± SEM.

^aValue versus control, ^bValue versus ME1, ^cValue versus ME2.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

TABLE 2 Mean $\pm$ SEM of per cent rate sperm motility, viability and DSP of experimental groups of rats, receiving different doses of methanol extract of *Asplenium dalhousiae*

Parameters	Treatments			
	Control	ME 1 (50 mg/kg)	ME 2 (100 mg/kg)	ME 3 (150 mg/kg)
Motility (%)	73.6 $\pm$ 3.33	62.8 $\pm$ 6.31	52.8 $\pm$ 3.31 ^{a*}	56.6 $\pm$ 5.17
Viability (%)	77.6 $\pm$ 4.42	62.8 $\pm$ 5.11	61.4 $\pm$ 1.07 ^{a**b*}	71.8 $\pm$ 2.22
Daily sperm production $\times 10^6$ /testis	2.95 $\pm$ 4.65	1.31 $\pm$ 6.04 ^{a***}	1.17 $\pm$ 5.47 ^{a***}	1.68 $\pm$ 1.10 ^{ac***b*}

Notes. Values are expressed as mean $\pm$ SEM.

^aValue versus control, ^bValue versus ME1, ^cValue versus ME 2.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

3.6 | Biochemical analysis

Methanol extract of *A. dalhousiae* treatment resulted in significant decrease in CAT ($p < 0.001$) and SOD ($p < 0.01$) activity within testis of all the treated groups when compared to control and among themselves. A substantial reduction ($p < 0.01$) in POD level was also evident in ME 3 group as compared to control and ME 1 groups. In all extract treated groups, a dose-dependent increase in ROS value was observed in comparison with control group. The value of TBARS was also amplified in all the treated groups as compared to control as well as with other treated groups (Table 3).

3.7 | Hormone analysis

Plasma testosterone levels were altogether ($p < 0.001$) brought down in ME 1 and ME 3 groups when compared with control group. Similarly, significantly reduced concentration of testosterone was noted in ME 2 group when comparison was made with control ($p < 0.01$; Table 3).

Reduction in LH and FSH levels was evident among all plant extract treated groups as compared to control group; however, this change was nonsignificant, excluding ME3 group where a significant change ($p < 0.05$) in FSH concentrations was evident (Table 3).

3.8 | Histomorphometric analysis

The histomorphological analysis for testicular tissues has been demonstrated in Table 4 and Figure 2.

3.8.1 | Testis morphology

On histological examination of testicular tissues of control group, closely spaced seminiferous tubules showing normal spermatogenesis were observed (Figure 2a). Mature spermatozoa were present in the narrow lumen. Leydig cells (oval, round or irregular shaped) along with large blood vessels were present in the interstitial spaces surrounding seminiferous tubules. ME1 (50 mg/kg) treatment group represented disorganised seminiferous tubules with increased diameter and wider lumen space. Because of loose germ cells, concentrated intracellular spaces in germinal epithelium were observed (Figure 2b). In ME2 group (100 mg/kg),

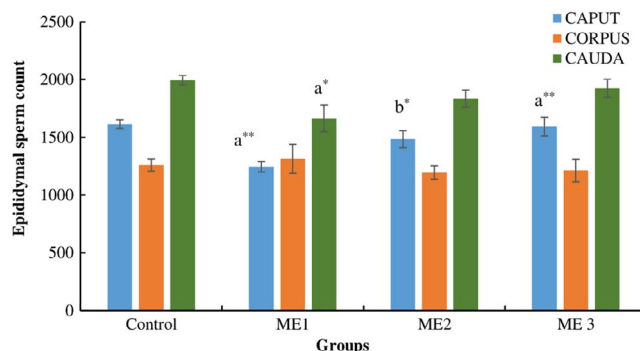


FIGURE 1 Effect of methanol leaf extract of *Asplenium dalhousiae* Hook. on epididymal sperm production in control and treated groups of adult male rats after 28 days of treatment

narrow lumen with increased epithelial height and further distorted Sertoli–Sertoli junction was evident (Figure 2c). Testicular sections of ME 3 group (150 mg/kg) showed quite unexpected results (Figure 2d) as compared to other treatment groups. Organised and arranged seminiferous tubules with minimal damage to epithelium and lumen flushing with spermatid were seen. The reasons for these changes in ME3 are not known and need further investigation.

A noteworthy decrease ($p < 0.01$) of tunica albuginea height was found in ME 1 group when contrasted with control group. No critical change in mean tunica albuginea height was seen in other two treatment (100 and 150 mg/kg) groups. Exceptionally significant expansion ($p < 0.001$) in tallness of tunica albuginea was seen among ME 2 and additionally ME 3 when contrasted with ME 1 group (Table 4).

3.8.2 | Stereological analysis

A significant reduction ($p < 0.01$) in number of spermatogonia was observed in all ME treated groups as compared to control. No significant difference in number of primary spermatocytes was seen in ME 1 group. However, significant reduction was noted in ME 2 ($p < 0.01$) and ME 3 ($p < 0.05$) groups in comparison with control group.

Highly significant decrease ($p < 0.001$) was noted in the number of secondary spermatocytes in all the plant extract treated groups (ME 1, ME 2 and ME 3) when compared to control group. Similarly, number of secondary spermatocytes in ME 3 was also reduced

TABLE 3 Effect of methanol leaf extract of *Asplenium dalhousiae* on biochemical parameters of rat testis

Parameters/enzymes	Treatment			
	Control	ME 1 (50 mg/kg)	ME 2 (100 mg/kg)	ME 3 (150 mg/kg)
CAT (U/mg)	39.69 ± 3.28	8.48 ± 1.36 ^{***}	18.48 ± 1.54 ^{***b*}	18.94 ± 2.81 ^{***b*}
SOD (U/mg)	45.00 ± 2.42	29.42 ± 3.07 ^{***}	26.11 ± 3.76 ^{***}	38.39 ± 2.81
POD (nmole)	6.16 ± 0.50	6.45 ± 0.89	5.21 ± 0.54	3.25 ± 0.22 ^{ab**}
ROS (μmol/min)	0.89 ± 0.05	1.24 ± 0.22 ^{a*}	1.58 ± 0.19 ^{***}	1.28 ± 0.02 ^{***}
TBARS (nM/mg)	2.01 ± 0.41	2.35 ± 0.50	3.47 ± 0.42	4.59 ± 0.47 ^{ac***b*}
Testosterone (ng/ml)	4.08 ± 0.79	0.67 ± 0.25 ^{***}	1.86 ± 0.74 ^{***}	0.59 ± 0.41 ^{***}
LH (IU)	0.66 ± 0.19	1.16 ± 0.21	1.09 ± 0.16	0.80 ± 0.21
FSH (IU)	1.63 ± 0.24	1.38 ± 0.15	1.16 ± 0.23	0.75 ± 0.12 ^{a*}

Notes. Values are expressed as mean ± SEM.

^aValue versus control, ^bValue versus ME1, ^cValue versus ME2.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

TABLE 4 Morphometry of testis and different number of cell types in seminiferous tubule in testis of rats exposed to different concentrations of *Asplenium dalhousiae* leaf extract

Morphometric parameters	Treatments			
	Control	ME 1 (50 mg/kg)	ME 2 (100 mg/kg)	ME 3 (150 mg/kg)
Tubule diameter (μm)	660.55 ± 25.46	494.09 ± 18.84 ^{a*}	491.72 ± 20.92 ^{a*}	516.80 ± 20.16 ^{a*}
Lumen diameter (μm)	522.18 ± 23.85	432.33 ± 28.85	409.86 ± 22.58 ^{ab*}	491.45 ± 30.44
Epithelial height(μm)	140.04 ± 27.80	60.15 ± 14.87 ^{***}	90.87 ± 23.78 ^{ab***}	35.70 ± 9.56 ^{ac***}
Tunica albugenia height (μm)	110.07 ± 6.77	76.91 ± 5.59 ^{a**}	112.40 ± 7.34 ^{b***}	129.95 ± 6.81 ^{b***}
Number of cell types				
Spermatogonia	86.6 ± 4.46	72.8 ± 2.74 ^{a*}	66.8 ± 1.60 ^{a**}	68 ± 3.12 ^{a**}
Primary spermatocytes	57 ± 3.68	46.8 ± 2.48	39.4 ± 2.92 ^{a**}	42.4 ± 3.8 ^{a*}
Secondary spermatocytes	50.6 ± 2.32	34.8 ± 1.69 ^{***}	28.4 ± 2.30 ^{***}	24.4 ± 1.73 ^{***b*}
Spermatids	108.6 ± 5.37	100.6 ± 3.15	74 ± 2.71 ^{ab***}	73 ± 2.58 ^{ab***}

Notes. Values are expressed as mean ± SEM.

^aValue versus control, ^bValue versus ME1, ^cValue versus ME 2.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

($p < 0.001$) when compared to ME 1 group. A significant reduction ($p < 0.001$) in number of spermatids was observed in ME 2 and ME 3 groups when compared with control as well as with ME 1 group, while no change was seen among ME 1 and control group (Table 4).

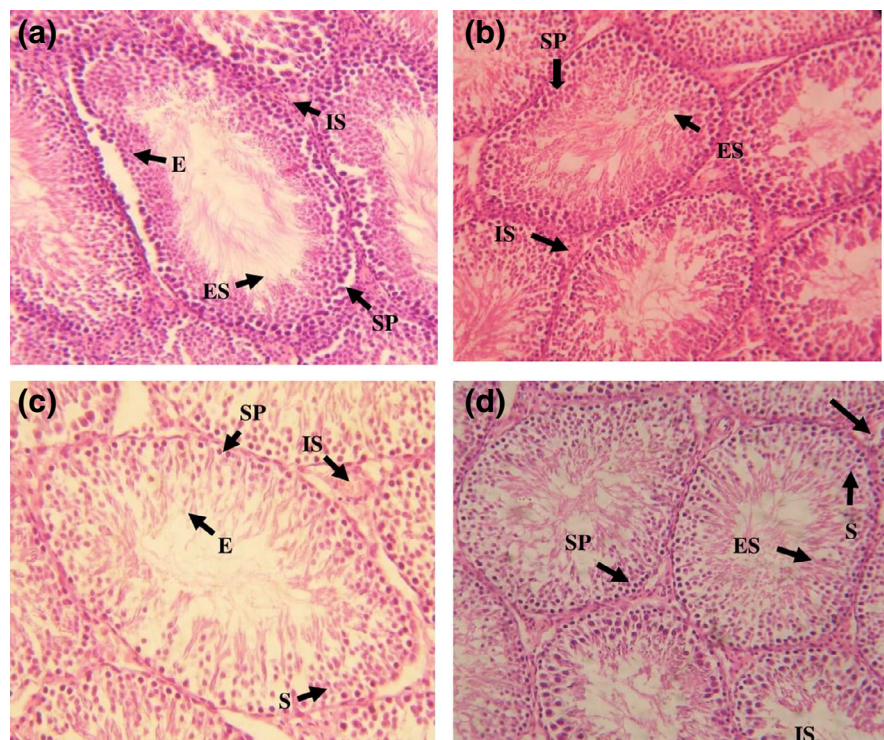
4 | DISCUSSION

This is the first study to investigate the possible antifertility properties of leaf extract of *A. dalhousiae* in male rats. The previous studies have focused on its various pharmacological properties (Chander et al., 2017; Khan et al., 2012); however, a review previously published by Reichstein (1981) had reported that this plant possesses contraceptive activity but no experimental work has been done so far to check this. The results of present study showed that treatment with methanol leaf extract of *A. dalhousiae* exerted hazardous and pronounced effects on fertility and reproduction of adult male rats through affecting sperm factors and disturbing/altering antioxidant status.

Our results have shown marked increase in body and testicular weight in rats exposed to different doses of *A. dalhousiae* leaf extract. An increase in weight of epididymis, seminal vesicle and prostate was also seen among all the treatment groups as compared to control.

Spermatogenesis is governed by two phases with respect to cell division, that is mitotic phase and meiotic phase. In present study, sperm motility and viability were reduced in all the treatment groups with different doses given, but remarkable reduction was only seen in 100 mg/kg treated group. Because this is the dose limit where the plant extract has exerted its peak activity. Whereas in 150 mg/kg dose treatment group, the results are nonsignificant as compared to control that might be because of desensitisation of sperm membrane receptors, thus making them irresistible to respond to further stress. The plant extract treated groups expressed not much difference in the sperm motility and viability because the extract affected the initial phases of spermatogenesis where spermatogonia proliferated via

FIGURE 2 Photomicrograph of seminiferous tubules. Dense tubules, lumen engorged with spermatids with normal germ cells shown in (a) Control; closely spaced seminiferous tubules (b) ME 1 group treated with 50 mg/kg of dose; showing tubules with empty lumen and degenerated epithelial layer with increased interstitial space, (c) ME 2 group treated with 100 mg/kg; showing narrow lumen, increased epithelial height and compact tubules with less interstitial space, (d) ME 3 group treated with 150 mg/kg of dose showing minimal damage to epithelium, lumen flushed with spermatid and less interstitial space. Magnification $\times 40$. E: epithelium; ES: elongated spermatids; IS: interstitial space; S: spermatocytes; SP: Spermatogonia



mitotic division and further differentiation occurred up till completion of meiosis I. As epididymis is involved in maturation of spermatozoa, the reduced sperm motility is also contributed to epididymal structural disruptions. Significant reduction in daily sperm production was evident among all the treated groups when compared to control, because extract might have not exerted its effects on later stages of spermatogenesis, that is meiosis II and spermiogenesis. Reduction in sperm count is probably due to antispermatogenic and antisteroidogenic effects of extract on testis. The reduced epididymal sperm count, here, is thought to be related to structural disruptions in caput and cauda epididymis induced by the plant extract.

Oxidative stress is induced by high production of free radicals, that is TBARs and ROS in cells and tissues. Dandekar, Nadkarni, Kulkarni, and Puneekar (2002) have reported that due to higher content of polyunsaturated fatty acids in the plasma membrane of spermatozoa, they are highly susceptible to elevated levels of ROS (Dandekar et al., 2002). With disturbed levels of antioxidants, lipid peroxidation is induced and, thus, disruption of lipid membranes occurs, resulting in reduced sperm motility and decreased spermatogenesis (Khaki, 2011). Our study has also shown dose-dependent increase in number of reactive oxygen species and induction of spermatogenic arrest due to actions of quercetin in testicular tissue.

CAT and SOD are two antioxidant enzymes with free radical scavenging property in male reproductive organs (Kumar, Murarka, Mishra, & Gautam, 2014). Our study has reported their decreased levels in dose-dependent manner in all treatment groups. POD is another important antioxidant enzyme involved in cell defence mechanism. Reduction in POD levels prevents the conversion of hydrogen peroxide into water through oxidation-reduction reaction, thus resulting in cell damage and death. The findings of present study have

shown that leaf extract of *A. dalhousiae* possess components that not only interfere with normal activities of antioxidant enzymes, but also reduce their concentrations in tissues thus disturbing the homeostatic balance. Due to presence of considerable level of polyphenolic compounds including flavonoids, flavonols, proanthocyanidines and kaempferol in the plant, it is suggested that plant parts can also be considered to show antioxidant activity in other cases (Ajay Singh Bisht, 2015).

Morphometric findings showed significant change in structures of testis between control and treatment groups. The decrease in tubular and luminal diameter of seminiferous tubule was evident in dose-dependent manner that is thought to be directly related with low production of testosterone as suggested by Mali, Ansari, and Chaturvedi (2002). Basement membrane impairments, degenerative changes and contraction of seminiferous tubule were evident. However, further studies at cellular and molecular level are required for better understanding of cell signalling mechanisms and pathways revealing deleterious actions of extract on testis. Though, it can be hypothesised that plant extract could cross the blood–testes barrier and directly exerts its actions on seminiferous tubules.

The interaction of germ cells with Sertoli cell is crucial for maintaining spermatogenesis, and any damage to Sertoli cells results in spermatogenic arrest (Jahan et al., 2009). In our study, the distortion in Sertoli–Sertoli cell junction was evident in all the treatment groups. This disorganisation is due to negative actions of plant extract on blood–testes barrier as also suggested by Sundar Rajan, Sarathchandiran, and Kadalmani (2013) in their respective findings.

The antifertility effect of *A. dalhousiae* leaf extract on testis has been studied by its actions on spermatogenic cycle. We found failure

of normal spermatogenesis as well as steroidogenesis in rats exposed to different doses of *A. dalhousiae* leaf extract for 28 consecutive days. Reduction in number of spermatocytes and spermatids was observed in all the treatment groups (50, 100 and 150 mg/kg). The reduction might be due to antispermato-genic activity of plant extract, although it has not been reported previously and needs further exploration. Our findings are in agreement with Sundar Rajan et al. (2013), where animals treated with different doses of herbal oral contraceptive suspension containing methanol extracts of plants *Carica papaya* leaves and *Capparis aphylla* aerial part were used and it disturbed spermatogenesis by reducing spermatocytes number (Sundar Rajan et al., 2013).

Change in LH and FSH level was seen among all plant extract treated group as compared to control; however, the change was non-significant. A significant reduction in the plasma testosterone level was also observed. Reduced testosterone production is directly linked to reduced Leydig cell number, eventually leading to spermatogenic arrest. These findings suggest that *A. dalhousiae* extract might have affected gonadotropin and androgen synthesis by interfering with hypothalamus–pituitary–gonadal axis.

5 | CONCLUSION

It is concluded from present study that exposure to *A. dalhousiae* leaf extract exerts antifertility effects in adult male rats by causing significant decrease in sperm motility, viability and daily sperm production. Reduced levels of antioxidants and plasma testosterone levels eventually result in spermatogenic arrest. These effects are thought to be mediated by oxidative stress induced in testicular tissues. Because of aforementioned physiological effects of extract on testes, it is suggested that *A. dalhousiae* leaves can be utilised in the development of antifertility drugs for men.

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CONFLICT OF INTEREST

The authors report no declarations of interest.

AUTHORS' CONTRIBUTIONS

The plant was collected and identified by Dr Mushtaq and extract was prepared by Wajid Zaman. Sarwat Jahan led the experimental design and approved study. Mehwish David conceived the study, performed the experimental work, analysed the results and wrote the paper. Qurat ul Ain performed initial part of the study and helped in compiling the results.

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